

NyayaMitra AI

Updated Main Dev Task Allocation Plan (with SQL Database)

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1. Overview of Database Architecture Update

To build a production-ready application, we have integrated a relational database (SQLAlchemy + PyMySQL). The backend will support two database engines: SQLite (the local file default requiring zero setup for judges) and MySQL (configured in environment variables for production storage). This ensures the prototype is easily shareable.

2. Team Structure & Focus Areas

Member 1 (Backend Dev)	Focus: FastAPI setup, SQL Models, local DB tables, RAG retriever API, chat endpoint.
Member 2 (Frontend Dev)	Focus: Vite App setup, Drag & drop zones, Dual-Pane shell, notice timelines, checklist tabs
Member 3 (AI Dev)	Focus: Gemini SDK setup, visual OCR pipeline, structured output validations, Telugu trans
Member 4 (QA & Speech)	Focus: Mock notice preparation, Pytest suites, Browser TTS playback controllers, calendar

3. Daily Integration Deliverables Checklist

Day 1: Platform Foundation and Database Skeleton

- **Goal:** Uvicorn backend serving health checks and React-Vite UI shell displaying dashboard skeleton.
- **Database:** SQLAlchemy ORM tables initialized successfully in local SQLite nyayamitra.db on server boot.

Day 2: Multimodal Ingestion and RAG Grounding

- **Goal:** Gemini multimodal OCR extracts notice metadata and RAG index matches applicable legal codes.
- **Database:** Extracted analysis parameters successfully written to the database documents table.

Day 3: Contextual Q&A Chat and Telugu Translation

- **Goal:** Mitra chatbot console functional. Summaries and checklists translated dynamically to Telugu on request.
- **Database:** Every user message and chatbot response logged in chat_logs relational database table.

Day 4: Speech Synthesizer, Polishing and Pitch Deck

- **Goal:** Speech audio player working in Telugu and English. Dynamic .ics calendar event downloading.
- **Presentation:** Setup README guides and slides outline for hackathon demo.